

A Counterexample to the Initial-Segment Numerical-Index Problem

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Source Problem

The source paper is:

M. Martin, J. Meri, M. Popov, and B. Randrianantoanina, *Numerical index of absolute sums of Banach spaces*, arXiv:1003.3269.

Problem 5.3 asks whether, if Z has a monotone basis $(e_m)_{m \geq 1}$ and

$$X_m = \text{span}\{e_k : 1 \leq k \leq m\},$$

then

$$n(Z) = \limsup_{m \rightarrow \infty} n(X_m).$$

The parenthetical version asks whether the assertion might remain true under stronger hypotheses such as a one-unconditional, one-symmetric basis.

$$n(X) \geq \limsup_{i \in I} [n(e_i) \|e_i\|] = \limsup_{i \in I} n(e_i) \lim_{i \in I} \|e_i\| = \limsup_{i \in I} n(e_i).$$

The result follows by just taking $\varepsilon \downarrow 0$. □

The easiest particular case of the above result is to Banach spaces with a monotone basis (i.e. a basis whose basic constant is 1).

Corollary 5.2. *Let Z be a Banach space with a monotone basis (e_m) and for each $m \in \mathbb{N}$, let $X_m = \text{span}\{e_k : 1 \leq k \leq m\}$. Then*

$$n(X) \geq \limsup_{m \rightarrow \infty} n(X_m).$$

It is known, see Section 6, that if $X = \ell_p$ then the inequality above is actually an equality, but we do not know whether the same is true in any other type of spaces.

Problem 5.3. Let Z be a Banach space with a monotone (or even one-unconditional, one-symmetric) basis $\{e_m\}_{m \in \mathbb{N}}$, and let $X_m = \text{span}\{e_k : 1 \leq k \leq m\}$. Is it true that $n(X) = \limsup_{m \rightarrow \infty} n(X_m)$?

Next examples show that the inequality in Theorem 5.1 may be strict.

Examples 5.4.

(a) **REAL CASE:** Let X be the three-dimensional real space given in Example 3.2 such that

$$n(X) > n(P_2(X)) = 0,$$

where $P_2(X)$ is the subspace of X spanned by the two first coordinates. Now, consider the space $Z = \ell_1(X)$ and for each $m \in \mathbb{N}$, we consider the subspace

$$\begin{aligned} Z_m &= \{x = (x_k) \in Z : x_{m+1}(3) = 0, x_k = 0 \ \forall k \geq m+2\} \\ &\equiv X \oplus_1 \cdots \oplus_1^m X \oplus_1 P_2(X). \end{aligned}$$

Then, we have $n(Z) = n(X) > 0$ and $n(Z_m) = n(P_2(X)) = 0$ by Corollary 2.6. Observe that

Result

Theorem 1. *Problem 5.3 has a negative answer for monotone bases. In fact, there is a real Banach space Z with a one-unconditional monotone basis $(f_m)_{m \geq 1}$ such that, for*

$$F_m = \text{span}\{f_1, \dots, f_m\},$$

one has

$$n(Z) > 0, \quad n(F_m) = 0 \quad (m \geq 2).$$

Consequently,

$$n(Z) > \limsup_{m \rightarrow \infty} n(F_m).$$

Proof Intuition

The source paper already contains almost all of the construction. It gives a three-dimensional real space X with a one-unconditional, one-symmetric basis (u, v, w) , positive numerical index, and with $\text{span}\{u, v\}$ isometric to the real Hilbert plane, hence numerical index zero. The space $Z = \ell_1(X)$ still has positive numerical index. The trick is only to permute the natural basis of Z : keep opening a new block before closing the previous one. Then every sufficiently large initial segment has a whole copy of the two-dimensional Hilbert plane as an ℓ_1 -summand, so its numerical index is forced down to zero.

Proof

Let X be the real three-dimensional space from Example 3.2 of the source paper:

$$\|(a, b, c)\| = \max \left\{ \sqrt{a^2 + b^2}, \sqrt{a^2 + c^2}, \sqrt{b^2 + c^2} \right\}.$$

The usual basis (u, v, w) is one-unconditional and one-symmetric. The source paper proves

$$n(X) > 0$$

and observes that

$$H := \text{span}\{u, v\}$$

is isometric to the real two-dimensional Hilbert space. Hence $n(H) = 0$.

Set

$$Z = \ell_1(X).$$

For the j -th copy of X , write its coordinate basis as

$$u_j, \quad v_j, \quad w_j.$$

By Corollary 2.6 of the source paper, the numerical index of an ℓ_1 -sum is the infimum of the numerical indices of the summands, so

$$n(Z) = n(\ell_1(X)) = n(X) > 0.$$

Now order the coordinate vectors of Z as follows:

$$f_1 = u_1, \quad f_2 = v_1,$$

and, for $k \geq 1$,

$$f_{3k} = u_{k+1}, \quad f_{3k+1} = v_{k+1}, \quad f_{3k+2} = w_k.$$

Thus the sequence begins

$$u_1, v_1, u_2, v_2, w_1, u_3, v_3, w_2, u_4, v_4, w_3, \dots$$

This is a permutation of the natural one-unconditional basis of $\ell_1(X)$, hence it is again a one-unconditional monotone basis.

Fix $m \geq 2$, and let

$$F_m = \text{span}\{f_1, \dots, f_m\}.$$

We claim that F_m contains, as an ℓ_1 -summand, a copy of

$$H_j := \text{span}\{u_j, v_j\}$$

for some j , while $w_j \notin F_m$. Indeed, after u_j and v_j are listed, the vector w_j is not listed until after u_{j+1} and v_{j+1} . At every initial stage $m \geq 2$, at least one currently open block has exactly its first two coordinates present and its third coordinate absent.

Since the norm on $Z = \ell_1(X)$ is the sum of the block norms and the basis of X is one-unconditional, each F_m is an ℓ_1 -sum of finitely many coordinate subspaces of copies of X . One of those summands is the Hilbert plane H_j , so Corollary 2.6 of the source paper gives

$$n(F_m) = \min\{\text{numerical indices of the nonzero block summands of } F_m\} \leq n(H_j) = 0.$$

As numerical indices are nonnegative, $n(F_m) = 0$ for every $m \geq 2$. Therefore

$$\limsup_{m \rightarrow \infty} n(F_m) = 0 < n(Z),$$

which disproves the asserted equality. □

Complex Variant

The same reordering also gives a complex one-unconditional counterexample, although not with zero as the limiting value. Use Example 3.3 of the source paper: a five-dimensional complex space X with one-unconditional basis,

$$n(X) = 1,$$

and whose first four-coordinate subspace $P_4(X)$ satisfies

$$n(P_4(X)) < 1.$$

In $Z = \ell_1(X)$, list the first four coordinates of block $j+1$ before listing the fifth coordinate of block j . Then every sufficiently large initial segment has a $P_4(X)$ summand, hence its numerical index is at most $n(P_4(X)) < 1$, while

$$n(Z) = n(\ell_1(X)) = n(X) = 1.$$

Thus the complex one-unconditional version is also false.

Limitations

The construction uses a one-unconditional monotone basis, but the permutation of coordinates destroys the global one-symmetry of the natural block basis. It therefore does not settle the stronger one-symmetric variant mentioned in Problem 5.3.

Novelty Check

On 2026-06-28, I searched the web for exact and close variants of the problem:

`"Numerical index of absolute sums of Banach spaces" "monotone" "basis" "limsup",`

`"n(X) = limsup" "numerical index" "monotone basis",`

and

`"numerical index" "dense increasing" "one-complemented subspaces".`

The only exact match found was the original arXiv paper. No later explicit answer to Problem 5.3 was found in this bounded search.

References

References

- [1] M. Martin, J. Meri, M. Popov, and B. Randrianantoanina, *Numerical index of absolute sums of Banach spaces*, arXiv:1003.3269.